



K_0 -Ring of Spanning Line Configurations and Fubini Word Combinatorics

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Background: The Coinvariant Algebra and the Flag Variety

Symmetric polynomials are invariants of $\mathcal{S}_n \curvearrowright \mathbb{Z}[x_1, \dots, x_n]$, with a basis given by the elementary symmetric polynomials e_i . The *coinvariant algebra* R_n is the quotient ring

$$R_n := \frac{\mathbb{Z}[x_1, x_2, \dots, x_n]}{(e_1, e_2, \dots, e_n)}.$$

Let $\mathbb{F}$ be a base field of characteristic not 2. A *complete flag* $E_\bullet$ in $\mathbb{F}^n$ is a nested sequence of subspaces

$$0 = E_0 \subseteq E_1 \subseteq \dots \subseteq E_n = \mathbb{F}^n$$

where E_i is a vector subspace of dimension i . The *flag variety* Fl_n is the parameter space of complete flags in $\mathbb{F}^n$. Equivalently, $Fl_n \cong GL_n/B$. The geometry of Fl_n is heavily intertwined with the coinvariant algebra and $\mathcal{S}_n$ -combinatorics [BGP25]. There is a *cell decomposition* of Fl_n given by *Schubert cells* C_ω , indexed by permutations $\omega \in \mathcal{S}_n$. A *Schubert variety* X_ω is the closure of C_ω .

A Few Cohomology Theories

Let X be a smooth $\mathbb{F}$ -scheme and VB_X be the exact category of vector bundles over X . The *Grothendieck group* of X is the (ungraded) ring

$$K_0(X) := \frac{\mathbb{Z} \cdot \mathrm{Iso} \, \mathrm{VB}_X}{[\mathcal{E}] = [\mathcal{E}'] + [\mathcal{E}''] \text{ whenever } \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \text{ is S.E.S.}}.$$

The *Chow ring* of X is the graded ring

$$CH^\bullet(X) := \frac{\mathbb{Z} \cdot \{ \text{subvarieties } Z \subseteq X \}}{\text{rational equivalence}}.$$

Since $X(\mathbb{C})$ is a smooth complex manifold, it has *integral cohomology* $H^\bullet(X(\mathbb{C}); \mathbb{Z})$.

The Borel Presentation [Bor53; Dem73]

There are ring isomorphisms

$$K_0(Fl_n) \cong CH^\bullet(Fl_n) \cong H^\bullet(Fl_n(\mathbb{C}); \mathbb{Z}) \cong R_n. \quad (1)$$

The isomorphism $K_0(Fl_n) \cong CH^\bullet(Fl_n)$ is *exceptional* in the sense of [Lar+24], given by $1 - [\mathcal{L}_i^\vee] \mapsto c_1(\mathcal{L}_i)$. The second $CH^\bullet(Fl_n) \cong H^\bullet(Fl_n(\mathbb{C}); \mathbb{Z})$ is a graded isomorphism given by the cycle class map.

Schubert and Grothendieck Polynomials

The classes $[X_\omega] \in CH^{\mathrm{inv}(\omega)}(Fl_n) \cong H^{2\mathrm{inv}(\omega)}(Fl_n(\mathbb{C}); \mathbb{Z})$ can be identified with *Schubert polynomials* $\mathfrak{S}_\omega(x_1, \dots, x_n) \in R_n$. The classes $[\mathcal{O}_{X_\omega}] \in K_0(Fl_n)$ can be identified with *Grothendieck polynomials* $\mathfrak{G}_\omega(x_1, \dots, x_n) \in R_n$.

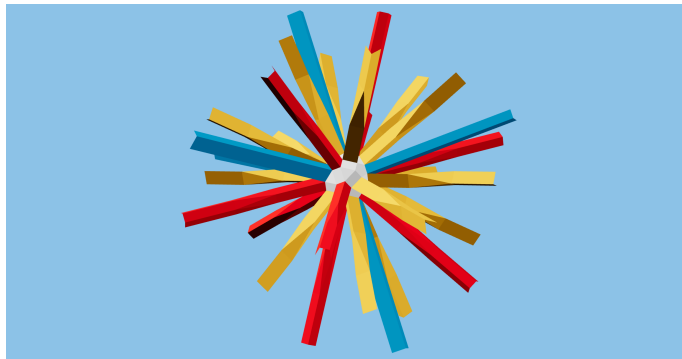
Examples $\mathfrak{S}_{12354} = x_1 + x_2 + x_3 + x_4 = e_1(x_1, \dots, x_4), \quad \mathfrak{G}_{12354} = e_1 - e_2 + e_3 - e_4.$

Ordered Set Partitions, aka Fubini Words

Fix $1 \leq k \leq n$ throughout. A *word* is a function $w : [n] \rightarrow [k]$, written as $w_1 \dots w_n$. A word w is a *Fubini word* if every letter appears at least once.

$(n, k) = (5, 3)$ Words: 12121, 21221 Fubini words: 12131, 32123, 23312

Fubini words are in bijection with *ordered set partitions* of $[n]$ with k (non-empty) parts, where $w : [n] \rightarrow [k]$ maps to $(w^{-1}(1) \mid w^{-1}(2) \mid \dots \mid w^{-1}(k))$. If $k = n$, Fubini words are permutations in $\mathcal{S}_n$. The number of ordered set partitions of $[n]$ with k parts is called *Fubini number* (OEIS A000670), given by $\# \text{Fubini}(n, k) = k! \cdot \mathrm{Stir}(n, k)$.



Generalized Coinvariant Algebra & Spanning Line Configurations

Haglund-Rhoades-Shimozono [HRS18] first studied the *generalized coinvariant algebra*

$$R_{n,k} := \frac{\mathbb{Z}[x_1, x_2, \dots, x_n]}{(x_1^k, x_2^k, \dots, x_n^k, e_{n-k+1}, e_{n-k+2}, \dots, e_n)}.$$

Pawłowski-Rhoades [PR19] constructed a variety whose cohomology is isomorphic to $R_{n,k}$. A *spanning line configuration* is an ordered n -tuple $\ell_\bullet = (\ell_1, \ell_2, \dots, \ell_n)$, where each ℓ_i is a one-dimensional linear subspace of $\mathbb{F}^k$ and $\ell_1 + \dots + \ell_n = \mathbb{F}^k$. The *variety of spanning line configurations* $X_{n,k}$ is the parameter space for all such $\ell_\bullet$. The variety $X_{n,k}$ is quasiprojective and can be identified as an open set of $(\mathbb{P}^{k-1})^n$. Pawłowski-Rhoades constructed an alternative cell decomposition of $(\mathbb{P}^{k-1})^n$ with *Pawłowski-Rhoades cells* indexed by *words* which descends to $X_{n,k}$ with cells indexed by *Fubini words*. They furthermore showed

$$H^\bullet(X_{n,k}(\mathbb{C}); \mathbb{Z}) \cong R_{n,k}$$

using *Fulton's degeneracy loci formula*. One goal of this paper is to extend the above isomorphism to an analogue of (1).

Theorem 1

There are ring isomorphisms

$$K_0(X_{n,k}) \cong CH^\bullet(X_{n,k}) \cong H^\bullet(X_{n,k}(\mathbb{C}); \mathbb{Z}) \cong R_{n,k}. \quad (2)$$

The isomorphism $K_0(X_{n,k}) \cong CH^\bullet(X_{n,k})$ is given by $1 - [\mathcal{L}_i^\vee] \mapsto c_1(\mathcal{L}_i)$. The second $CH^\bullet(X_{n,k}) \cong H^\bullet(X_{n,k}(\mathbb{C}); \mathbb{Z})$ is a graded isomorphism given by the cycle class map.

Proof Ingredients

Fulton-Lascoux Degeneracy Loci Formula [FL94]

Let Ω_ω be the degeneracy locus of a map of flagged vector bundles $\mathcal{E}_\bullet \rightarrow \mathcal{F}_\bullet$ of shape $\omega \in \mathcal{S}_n$. The class $[\mathcal{O}_{X_\omega}] \in K_0(X)$ is represented by a *Grothendieck polynomial* $\mathfrak{G}_\omega$. See [Theorem 3](#).

K_0 -Ring of an Open Complement (Theorem 2)

Let X be a cellular variety and Z be a union of closures of cells. Then, $K_0(X - Z) \cong K_0(X)/I(Z)$.

Schubert and Grothendieck Polynomials of Words in $[k]^n$

Pawłowski-Rhoades defined the *convexification*, *associated permutation*, and *standardization* of a word $w \in [k]^n$.

Examples $w = \textcolor{teal}{2}\textcolor{teal}{4}12\textcolor{red}{3}43$ $\mathrm{conv}(w) = \textcolor{teal}{2}\textcolor{teal}{2}\textcolor{teal}{4}14\textcolor{red}{3}3$ $\sigma(w) = 1423657$ $\mathrm{std}(\mathrm{conv}(w)) = \textcolor{teal}{2}\textcolor{teal}{5}\textcolor{teal}{4}67\textcolor{red}{3}8\textcolor{teal}{1}$

Definition [PR19] The *Schubert polynomial* of w is

$$\mathfrak{S}_w(x_1, \dots, x_n) := \sigma(w)^{-1} \cdot \mathfrak{S}_{\mathrm{std}(\mathrm{conv}(w))}(x_1, \dots, x_n).$$

The Schubert polynomials $\{\mathfrak{S}_w \mid w \in \text{Fubini}(n, k)\}$ form an additive basis of $R_{n,k}$.

Definition We define the *Grothendieck polynomial* of w as

$$\mathfrak{G}_w(x_1, \dots, x_n) := \sigma(w)^{-1} \cdot \mathfrak{G}_{\mathrm{std}(\mathrm{conv}(w))}(x_1, \dots, x_n).$$

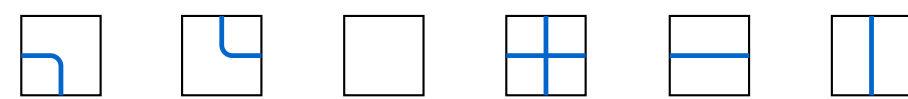
Theorem 3

Let $C_w \subseteq (\mathbb{P}^{k-1})^n$ be the Pawłowski-Rhoades cell of a word $w \in [k]^n$ and X_w be its closure. Under the isomorphism $K_0((\mathbb{P}^{k-1})^n) \cong \mathbb{Z}[x_1, \dots, x_n]/(x_1^k, \dots, x_n^k)$, the class $[\mathcal{O}_{X_w}]$ is mapped to the Grothendieck polynomial $\mathfrak{G}_w(x_1, \dots, x_n)$. Consequently, $\{\mathfrak{G}_w \mid w \in \text{Fubini}(n, k)\}$ forms another additive basis of $R_{n,k}$.

Pipe Dream Combinatorics

See [BGP25] for standard facts and references. A *classical pipe dream* P is a tiling of the $\mathbb{Z}_+ \times \mathbb{Z}_+$ grid by $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}$ and $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}$ -tiles. One obtains a permutation by chasing the pipes. For $\omega \in \mathcal{S}_n$, we have $\mathfrak{S}_\omega(x_1, \dots, x_n) = \sum_{P \in \mathrm{PD}(\omega)} x^P$, and one can recover the Grothendieck polynomial similarly.

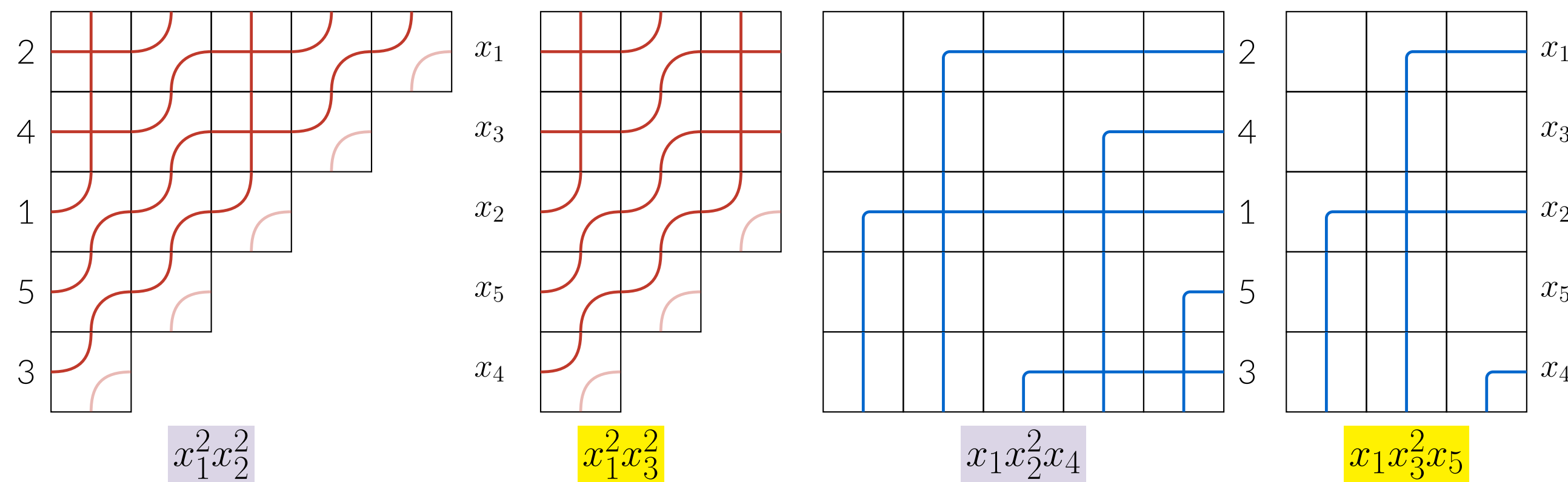
A *bumpless pipe dream (BPD)* B is a tiling of the grid $\mathbb{Z}_+ \times \mathbb{Z}_+$ in matrix coordinates by the following six tiles



that route pipes monotonically in the northeast direction. Schubert and Grothendieck polynomials are also weight-generating functions for BPDs.

Definition We define the *classical / bumpless pipedream of a word* $w \in [k]^n$ to be the leftmost k columns of a PD/BPD of the permutation $\mathrm{std}(\mathrm{conv}(w))$ with rows relabeled according to $\sigma(w)$.

Examples Left to right: PD for 24153, PD for 21231, BPD for 24153, BPD for 21231.



The following theorem is completely analogous to classical results on pipe dreams of permutations [BB93; LLS21; Wei21].

Theorem 4

The Schubert and Grothendieck polynomials of $w \in [k]^n$ are monomial-weight generating functions for classical / bumpless pipe dreams of w , defined as above.

References

This poster is based on a paper in preparation titled *The Grothendieck Group of the Variety of Spanning Line Configurations* (Zeng, 2025+).

[BB93] N. Bergeron and S. Billey, "rc-graphs and Schubert polynomials", English. In: *Exp. Math.* 2.4 (1993), pp. 257–269. DOI: 10.1080/10586458.1993.10504567.
[BGP25] S. C. Billey, Y. Gao, and B. Pawłowski, "Introduction to the Cohomology of the Flag Variety". In: *Handbook of Combinatorial Algebraic Geometry*. Ed. by M. P. et al. In preparation. unknown, 2025.
[Bor53] A. Borel, "Sur La Cohomologie des Espaces Fibrés Principaux et des Espaces Homogènes de Groupes de Lie Compacts". In: *Annals of Mathematics* 57 (1953), p. 115.
[Dem73] M. Demazure, "Invariants symétriques entiers des groupes de Weyl et torsion". In: *Invent. Math.* 21 (1973), pp. 287–301. DOI: 10.1007/BF01389390.
[FL94] W. Fulton and A. Lascoux, "A Pieri formula in the Grothendieck ring of a flag bundle". In: *Duke Mathematical Journal* 76.3 (1994), pp. 711–729. DOI: 10.1215/S0012-7094-94-07627-8.
[HRS18] J. Haglund, B. Rhoades, and M. Shimozono, "Ordered Set Partitions, Generalized Coinvariant Algebras, and the Delta Conjecture". In: *Advances in Mathematics* 329 (2018), pp. 851–915. DOI: 10.1016/j.aim.2018.01.028.
[LLS21] T. Lam, S. J. Lee, and M. Shimozono, "Back stable Schubert calculus". English. In: *Compos. Math.* 157.5 (2021), pp. 883–962. DOI: 10.1112/S0010437X21007028.
[Lar+24] M. Larson, S. Li, S. Payne, and N. Proudfoot, "K-rings of wonderful varieties and matroids". English. In: *Adv. Math.* 441 (2024). Id/No 109554, p. 43. DOI: 10.1016/j.aim.2024.109554.
[PR19] B. Pawłowski and B. Rhoades, "A Flag Variety for the Delta Conjecture". In: *Transactions of the American Mathematical Society* 372.11 (2019). Article electronically published on September 12, 2019, pp. 8195–8248. DOI: 10.1090/tran/7918.
[Wei21] A. Weigandt, "Bumpless Pipe Dreams and Alternating Sign Matrices". In: *Journal of Combinatorial Theory, Series A* 182 (Aug. 2021).